

Eskom MES applications

Input to the Consultative Environmental Panel Hearings – Medupi and Matimba

24 November 2022 (rev 1)

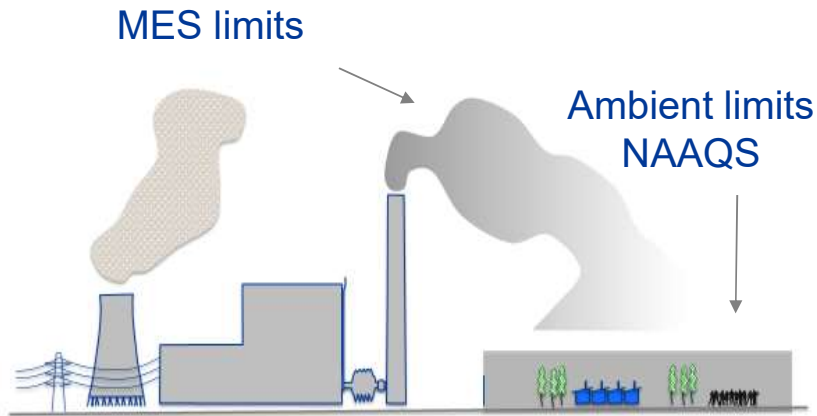


Objectives of today's discussions



- 1 **Talk about the Minimum Emission Standards (MES)**, and why Eskom has made an application to ask for relaxation for some aspects of them
- 2 **Discuss the impact of implementing the DFFE MES decision**, and why Eskom has appealed the decision
- 3 **Introduce Eskom's 2035 Strategy and Just Energy Transition plan**
- 4 **Show Eskom's Emission Reduction Plan 2022**

Minimum Emission Standards (MES) and ambient air quality



In settlements up to 50% of PM exceedances are associated with household fuel use

- The MES measures the limits set for the power station stacks.
- Ambient limits in terms of the National Ambient Air Quality Standards (NAAQS) measure public exposure at ground level.
- Eskom does contribute to emission loading in the region
- **Mpumalanga and Vaal:**
 - The PM ambient standard is exceeded in multiple hotspots. These exceedances are associated with low level sources such as fugitive dust and domestic coal burning.
 - SO₂ levels in general compliance but there are hotspots of exceedance and NO₂ is in compliance.
- **Waterberg:**
 - PM standard is exceed at times due to low level sources. NO₂ and SO₂ are in compliance with the NAAQS standard.
- The implementation of this emission reduction plan and the shutting down of stations will improve air quality from now to 2035.
- Low level sources (wind blown dust, veld burning, burning mines, household fuel use) which contribute to PM exceedances will not be addressed without integrated initiatives by Government and Industry. For example the Eskom air quality offset plan.

The MES regulations and the challenges

Legislation

- Existing plants which will be decommissioned before 2030 may apply for a once off suspension from new plant standards, they must have a detailed decommissioning schedule.
- If a plant is decommissioned after 2030, it must comply with the new plant standards within 5 years, by 2025 or shut down,
- No postponement of compliance timeframes or a suspension shall be granted for compliance with MES for existing plant standards.
- Alternative limits for a pollutant will be considered if there is: compliance with other pollutant standards at the station; if there is proof of investment in emission reduction on the pollutant at the station and; if there is material compliance with existing ambient standards.

19 February 2022

Challenges

- Most of the fleet more than 40 years old
- Stations not designed for retrofits and compliance has financial and other costs
- Legislation looks at each power station individually not across the fleet

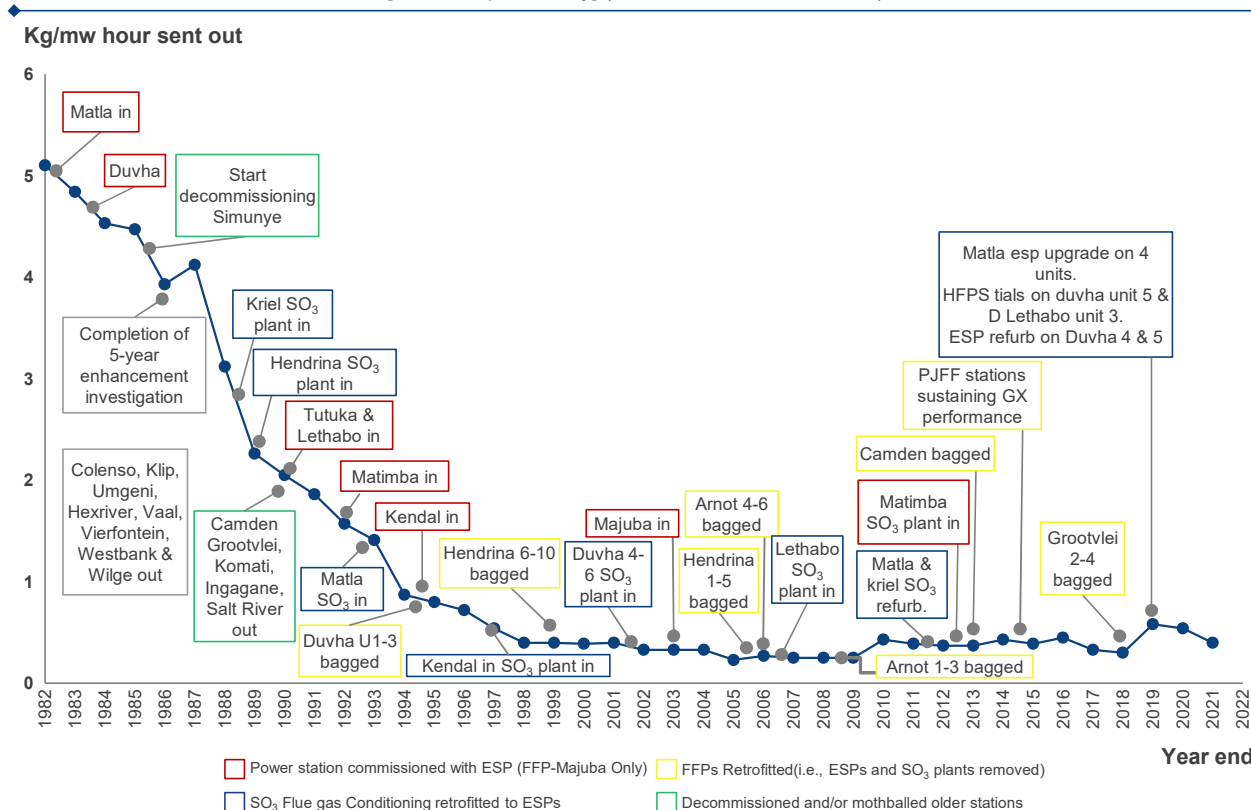


Eskom has made steady progress reducing emissions since the 1980's



Relative Emissions Roadmap-Particulates

[coal fired plants only] (Excl. Camden until FY2009)



- Since the 1980s, Eskom implemented a **phased approach**, prioritizing reduction of Particulate Matter (PM)
- Stations were **decommissioned**, or **retrofitted** with fabric filters, SO₃ plants, and improved electrostatic precipitators
- These actions resulted in a **reduction of PM emissions by >80%** from 1982 to 2021
- Emissions will reduce further over time

Note: From 2006, Years Represent Financial Years (April to March)

PM = Particulate Matter; CO₂ = Carbon Dioxide; SO₂ = Sulphur Dioxide; SO₃ = Sulphur Trioxide; NO₂ = Nitrogen Dioxide; ESP = Electrostatic precipitator; FFP = Fabric Filter Plant

The Eskom 2019 MES application described the financial, environmental and social cost of MES compliance



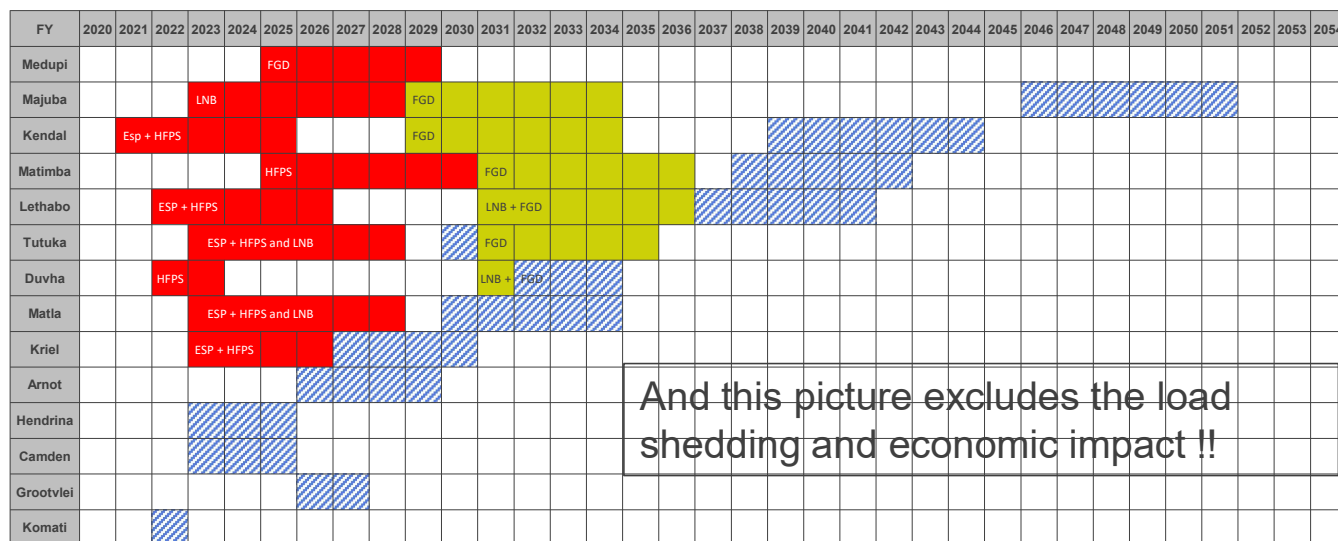
Implications		Full compliance with MES	Eskom's emission reduction plan (2019)
	SO ₂	Wet FGD at: Medupi, Majuba, Kendal, Matimba & Tutuka. Semi-dry FGD at: Duvha, Matla & Lethabo	Wet FGD at Medupi
	NO _x	LNBS at: Majuba, Matla, Tutuka, Lethabo, Duvha and Camden	LNBS at: Majuba, Matla, Tutuka and Camden
	PM	FFP at: Tutuka ESP Refurb/Upgrade at: Duvha, Kendal, Kriel, Lethabo, Matla and Tutuka	FFP at: Tutuka ESP Refurb/Upgrade at: Duvha, Kendal, Kriel, Lethabo, Matla and Tutuka
Water consumption increase		20% - (59 million m ³ /annum)	2% - (9.6 million m ³ /annum)
CAPEX retrofit & Offset OPEX (nominal)		R300 billion plus	R60 billion
Annual OPEX costs (2018 costs)		R5.9 billion	R900 million
Electricity tariff increase		7 to 10 %	2 to 3 %
Auxiliary power consumption increase		2 500 123 MWh/year ¹	400 192 MWh/year
CO ₂ emission increase (direct emissions from the FGD process only)		2.8-3.0 million tons/annum	435 000 tons/annum
Increase in coal consumption due to low NO _x burner retrofits		735 105 tons/annum	555 369 tons/annum
Waste (FGD by-product) production		9.7 million tons/annum ¹	2.7 million tons/annum
Sorbent Consumption		5.2 million tons/annum ¹	1.5 million tons/annum

- The Eskom plan reduces emissions over time
- Eskom's is not asking to be allowed to emit any more
- The Eskom plan reduces water and waste impacts
- The plan is practical and affordable to Eskom and South Africa and is in line with "**Sustainable development principles**"

Complying with the DFFE 2021 MES decision will require up to 14 years & >R300bn in capital



DFFE decision requires units to shutdown until the comply and necessary refits can only be completed between 2029 and 2036, shortly before or after most plants are decommissioned



And this picture excludes the load shedding and economic impact !!

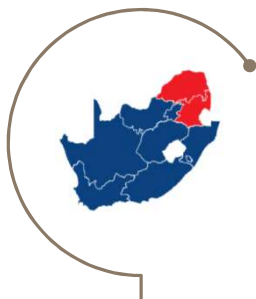
- Emission abatement retrofits/upgrades (Phase 1)
- Emission abatement retrofits/upgrades (Phase 2)
- ▨ Decommissioning (Eskom expected shutdown dates)

FFP = Fabric filter plant retrofit
 FGD = Flue gas desulphurisation retrofit
 LNB = Low NOx Burner installation/optimisation retrofit
 HFPS = High frequency power supply
 ESP = ESP, FGC, DHP Refurbishment/upgrade

Given the time frames, refitting most plants would be imprudent, constituting fruitless & wasteful expenditure

- 8 power stations require refit at a cost of R30 – R50bn per station (R300bn total)
- The '25 deadline is unachievable as it takes 7 – 10 years to build Flue Gas Desulphurization (FGD) plants
- Several stations **will shut down before** installation is complete (Matla, Duvha, Kriel)
- Others will **shutdown shortly after** the FGD is completed (Matimba, Lethabo, Kendal)
- Majuba and Matimba are in sparsely populated
- FGD will require additional 59 million cubic meters of water/annum.
- Additional 9.7 million tons of waste/annum produced.

Complying with the DFFE MES decision requires the premature shut down of power stations and would be catastrophic for Eskom & South Africa



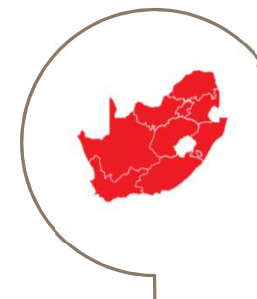
Direct impact on South Africa due to closure of power stations

- **~100 000 job losses** permanent job losses on closure of stations (direct & indirect)
- **18% & 33% negative impact to Limpopo & Mpumalanga GDP¹** respectively
- **R1.7 trillion cost to economy due to unserved energy**



Direct impact on Eskom's balance sheet (and govt. guarantees) due to loss of sales²

- Further **~R240bn operating shortfall** over 3 years (Eskom no longer a going concern)
- **Ballooning of debt burden to at least R640bn**, & credit rating downgrades
- **Potential default** on existing debt, & **call up of up to R350bn govt. guarantees**



Indirect impact on the economy due to generation shortfall when Eskom complies

- The high risk of a **national blackout** and economic catastrophe
- **Loadshedding intensifies up to 40X** current levels bringing the economy to a standstill
- **Further tariff increases** of up to **~170%** to pay for alternative replacement generation

1. Based on 2019 GDP and October Production Plan energy sent out, assuming 139c tariff across all stations. 2. Assuming 40% drop in sales
Source: Sources: <https://cer.org.za/wp-content/uploads/2018/02/Economic-impact-assessment-of-5-CFPSs.pdf>. Quarterly and Regional_Fourth_quarter 2020
<http://www.statssa.gov.za/?p=14171>

Eskom conducted extensive system energy modelling to find an optimal, comprehensive, long-term solution



Objective

Provide an **energy pathway up to 2050**, with a focus on the **next 14 years (to 2035)** given current constraints

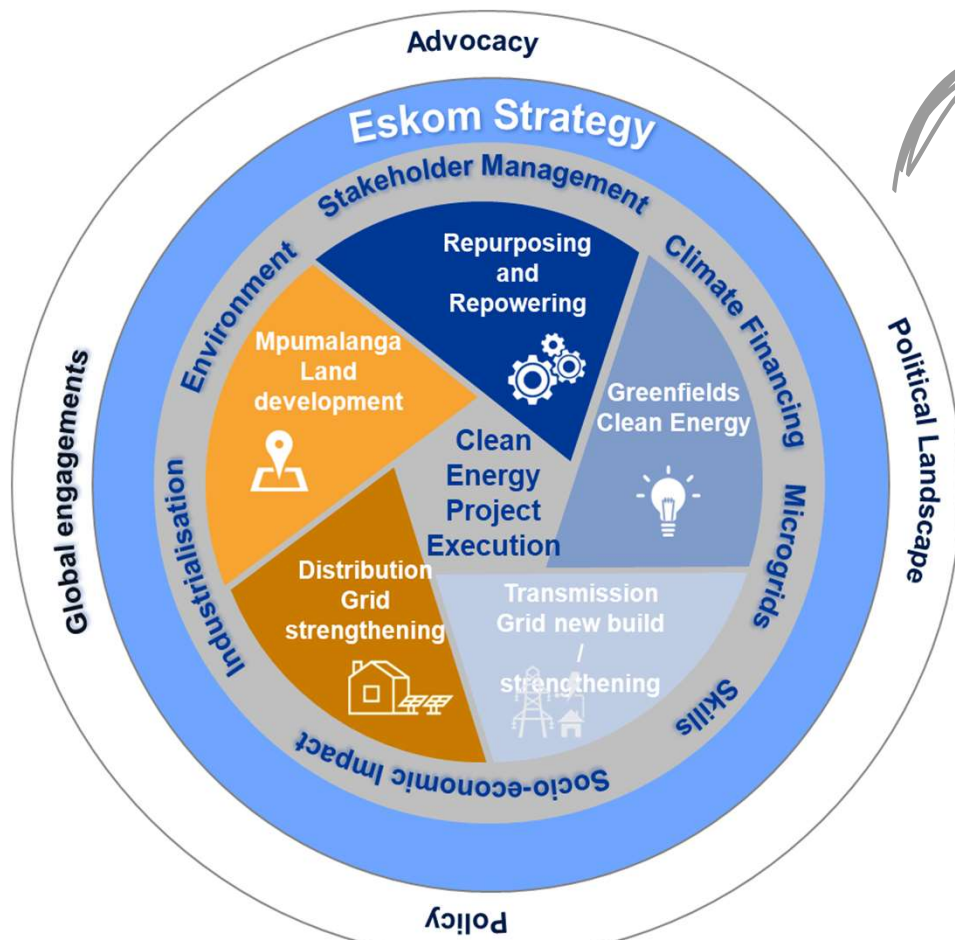
This study aimed to answer three key questions:

1. What is the **optimal future energy mix and capacity for SA** given practicality and the South African context?
2. What **additional capacity** should SA (incl. Eskom) build & how do we maximise the **socio-economic benefits** of a transition?
3. What are the impacts of an optimized shut down on **water use**, **emissions**, and **transmission & distribution** networks?

The results provided direction in terms of the following key strategic decisions:

- **Viability of the coal shutdown plan**, taking into account system requirements (**based on IRP**)
- **Alternative MES position**, optimising plant shutdown for a JET plan
- **Optimal build of new technologies** i.e., Renewables and Gas
- **5-year increments of energy mix** starting from 2025, 2035, 2040, up to 2050 Energy mix
- **Eskom 2035 plan developed**

To achieve this, Eskom has developed a clear & holistic JET strategy, including a pipeline of 100 projects



Examples of key projects

KOMATI

- **100 MW PV and 150 MW BESS**
- RFI for gas
- Opening of training centre
- **R 9 Billion loan obtained**

MAJUBA

- Land used for transformers
- **65 MW PV plant**

LETHABO

- **75 MW PV plant**

MICROGRIDS

- Fully functional assembly plant at Komati, with an ambition for 10 lines, employing **40 people per line**.
- Microgrids as EV charging stations

GAS

- **3000 MW South Coast Gas**
- **3000 MW Richards Bay Gas decision**

MPUMALANGA LAND

- Auctioning process completed
- Land leases signed
- Potential up to **4000 MW**

SERITI/ EXXARO MOU

- **~ 100 MW Matla / Kriel PV project decision**

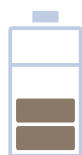
RSA JET Investment plan R 152 billion, 90% for electricity

By undergoing a JET and ERP 2022 we will reduce emissions while avoiding the impact a blunt MES implementation would have



Emissions

38% reduction in CO₂,
45% reduction in SO₂
emissions; 77% for PM



Generation capacity

22GW of coal gen.
replaced, & >7GW more
from new green energy



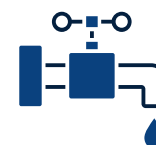
Jobs & the economy

>300 000 new net jobs
(after coal shutdowns);
reduced loadshedding



Transmission & Distribution

~15 500km of new
transmission &
distribution lines



Water scarcity

40bn litre reduction in
water consumption from
plant refits/shutdowns



JET impact ('35)

MES impact

Will immediately restrict
harmful emissions



Immediate loss of 16GW,
30GW loss by 2025



~100k job losses; stage
15 loadshedding



Added financial pressure
precludes investment



Requires >16bn more
litres of water per year



Note: PM = particulate matter; NOx = nitrogen oxide/dioxide/trioxide etc; SO₂ = sulphur dioxide
Source: GreenCape analysis (work in progress) WIP, RES4Africa studies; Eskom JET phase II modelling

Eskom Emission Reduction Plan 2022



Eskom 2035

Nine coal-fired power stations will be shut down by 2035, reducing the coal capacity by ~22 GW resulting in a total emission load reduction and enabling a Just Energy Transition

PM reduction projects

Kendal, Matimba, Lethabo,
Tutuka,
Duvha, Matla, Kriel

NOx reduction projects

Majuba, Lethabo, Tutuka

SO₂ reduction projects

Kusile and Medupi

PM, NOx and SO₂ optimization at all stations to meet requested limits

Air quality offsets

Highveld 35 000 house and waste solution in the Vaal Waterberg investigations and implementation

Budget total cost R 60 bn (nominal)
(Capex and Opex)

Assumptions

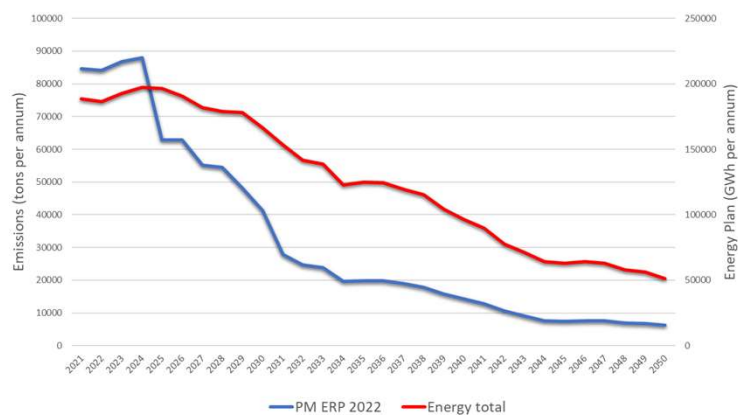
- Legal indulgence granted by DFFE to allow stations to operate whilst retrofits underway and above the existing MES limits.
- Funding for projects confirmed and delivery expedited
- Station emission performance optimised
- Capacity replacement projects delivered on time

¹ Based on Eskom 2035 scenario plan 2 c iv update 2 November 2022. Station shutdown as per IRP except Tutuka

Emissions decrease compared to 2021 45% SO₂, 55% NOx, 77% PM and 38% CO₂ emissions by 2035¹

Emission reduction over time under ERP 2022

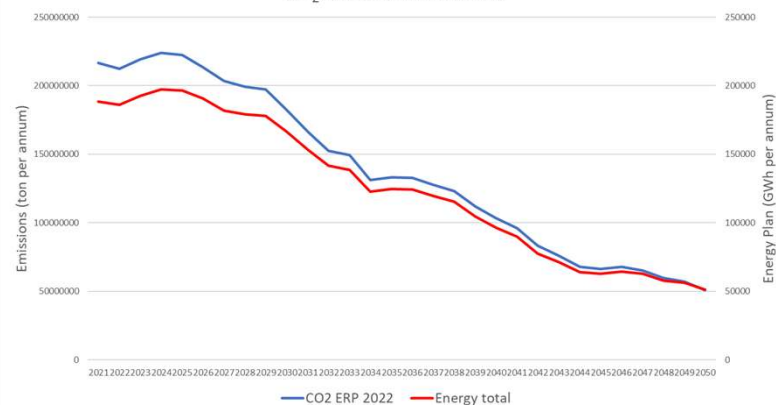
PM emissions over time



Emissions over time



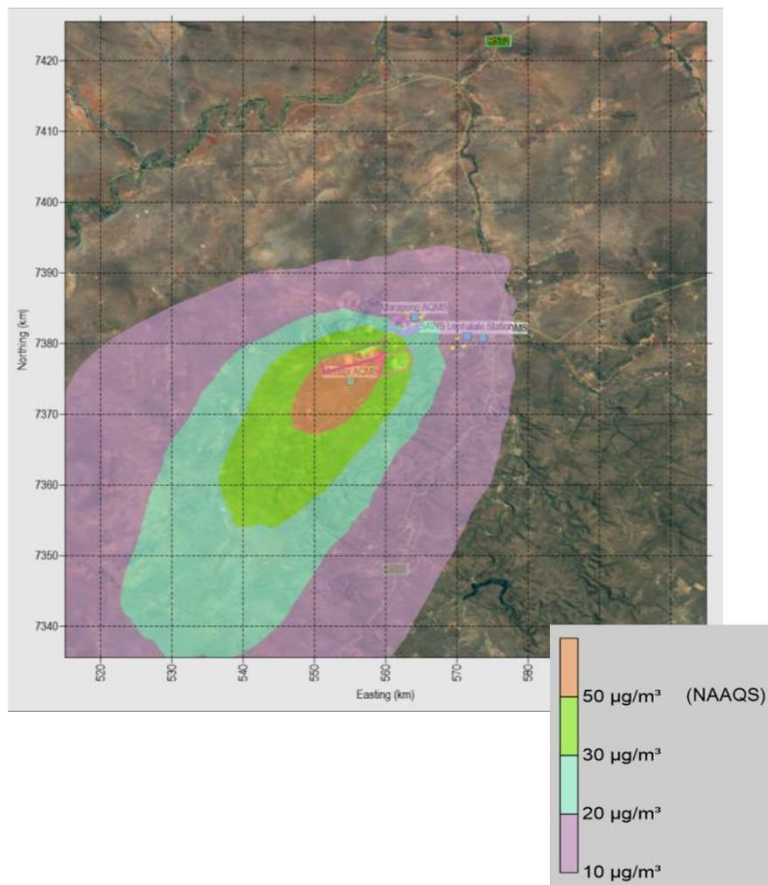
CO₂ emissions over time



Year/ Tons	2021	2025	2030	2035	Reduction percentage 2021-2035
PM	84 640	62 784	41 234	19 862	77%
NO _x	866 712	876 394	621 988	394 176	55%
SO ₂	1 837 991	1 850 154	1 427 506	1 015 973	45%
CO ₂	216 461 897	222 305 450	182 379 112	133 232 833	38%

Ambient air quality and health impacts – Waterberg Bojana priority area

Future baseline SO₂ annual



- Present ambient monitoring shows SO₂ and NO_x in compliance with NAAQS standard, PM exceedance due to low level sources observed.
- Dispersion modeling assuming both Medupi and Matimba operating shows continued PM, NO_x compliance.
- Annual average SO₂ shows compliance at all sensitive receptors with exception of Medupi monitoring station (next to station).
- Dispersion modeling predicts potential non-compliance to hourly and daily SO₂ standards but this is not shown in the monitoring data, so the model is over predicting.

Applications to be submitted and/or proposed requested alternative limits – 2019 & 2022 MES application

Station	Pollutant – new plant limit (existing plant limit)		
	PM – 50 (100)	NO _x – 750 (1100)	SO ₂ - 1000 (3500)
Coal-fuelled			
2019 application Matimba (Decom 2037-41)	None- Eskom will comply	OPTION 1 - Postponement OPTION 3 – Alternate monthly limit of 750 mg/Nm ³ from 2025 until decommissioning	OPTION 1 - Postponement OPTION 3 - Alternate limit of 4000 mg/Nm ³ monthly from 2020 until decommissioning
2022 application Matimba (Decom 2037-41)	⁹ Alternate monthly limit PM = 50 mg/Nm³ (monthly) until shutdown	⁹ Alternate monthly limit NO_x = 750 mg/Nm³ (monthly) until shutdown	¹⁰ Alternate monthly limit of SO₂ = 4000 mg/Nm³ (monthly) until shutdown
Medupi	None- Eskom will comply	None- Eskom will comply	OPTION 1 - Postponement OPTION 3 - Alternate limit of 4 000 mg/Nm ³ monthly from 2020 until 31 March 2030 (construction complete) then 1000 mg/Nm ³ until decommissioning.
Medupi Decom (2065-69)	None- Eskom will comply PM = 50 mg/Nm³	None- Eskom will comply NO_x = 750 mg/Nm³	¹¹ Alternate monthly limit of SO ₂ = 4000 mg/Nm ³ (monthly) until 2031 ¹¹ Alternate monthly limit of SO₂ = 1000 mg/Nm³ (monthly) from 2031 (retrofit complete) until shutdown

Matimba – PM and NO_x comply (monthly limit requested), SO₂ remain at existing levels

Medupi – PM and NO_x comply, SO₂ will comply once FGD installed 2031

Conclusions

- South Africa is at a critical time in terms of climate change, environmental and energy planning issues.
- Eskom's 2035 plan and Just Energy Transition plan are important for taking the country and the communities Eskom works with forward.
- Eskom believes that investing in JET and new “green” energies rather than spending money on old power stations which will be closing down soon is the most efficient, cost-effective use of Eskom and South Africa's limited financial resources.
- The Eskom Emission Reduction Plan and MES application provide a practical, affordable plan to reduce emissions over time, they are in-line with government plans and the principles of sustainable development and should be approved by the authorities.